

From Asymptotic Speedups to Practical Quantum Advantage

Guglielmo Mazzola



Outline

1. Constraints for practical quantum speed-up

- Input
- Output

2. Quantum-enhanced Monte Carlo

3. **Fully-quantum walks**

(Some) Challenges of quantum computing

Practical

- Digital quantum computers, still do not exist
- Current ones still noisy, small, etc..

(Some) Challenges of quantum computing

Practical

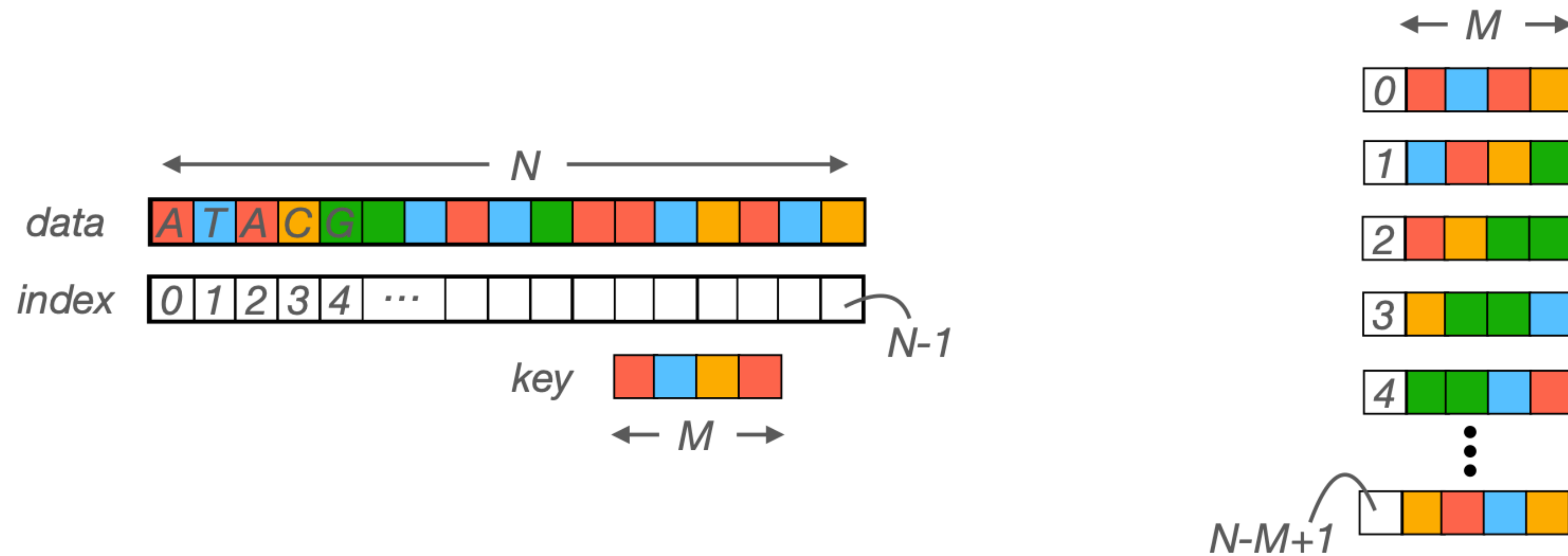
- Digital quantum computers, still do not exist
- Current ones still noisy, small, etc..

Conceptual

- Input: how do we insert the data?
- Output: how do we read the solution?

Input model: an example

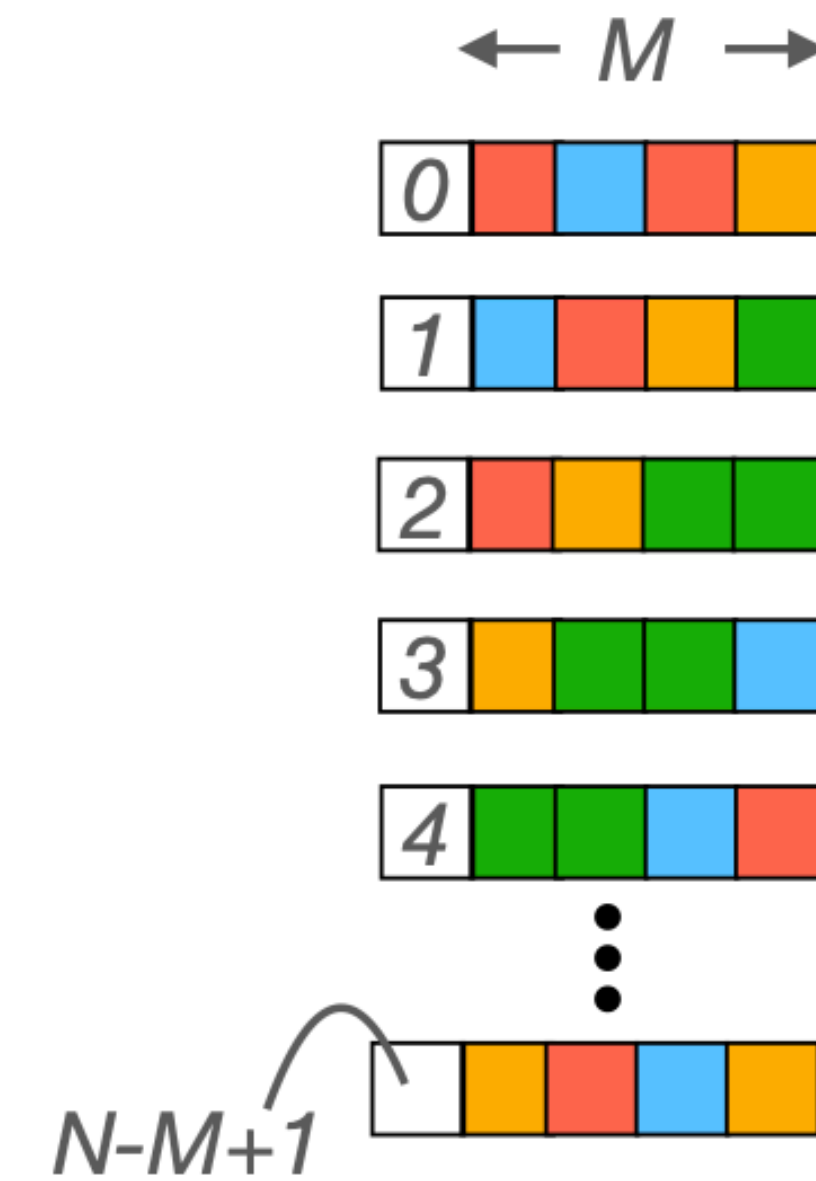
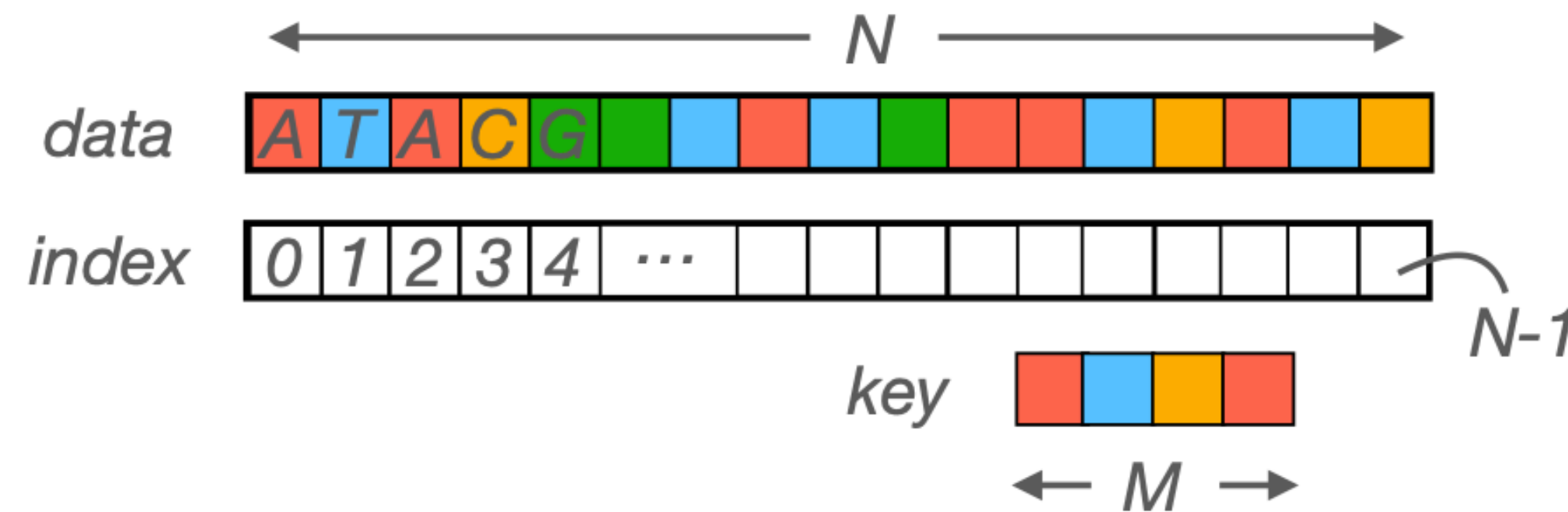
genome



Transformed
into a database
"phonebook"

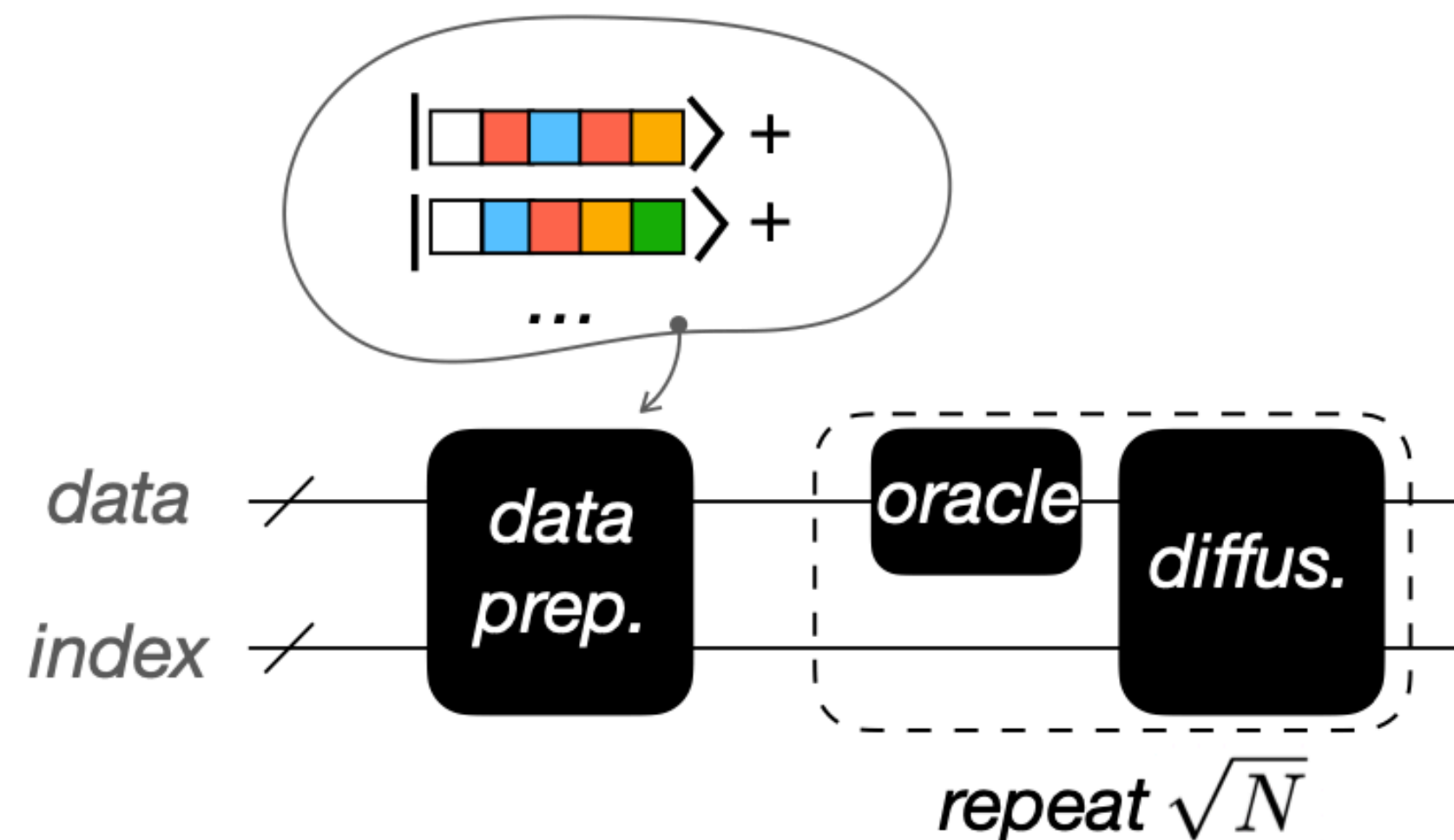
Input model: an example

genome



Transformed
into a database
"phonebook"

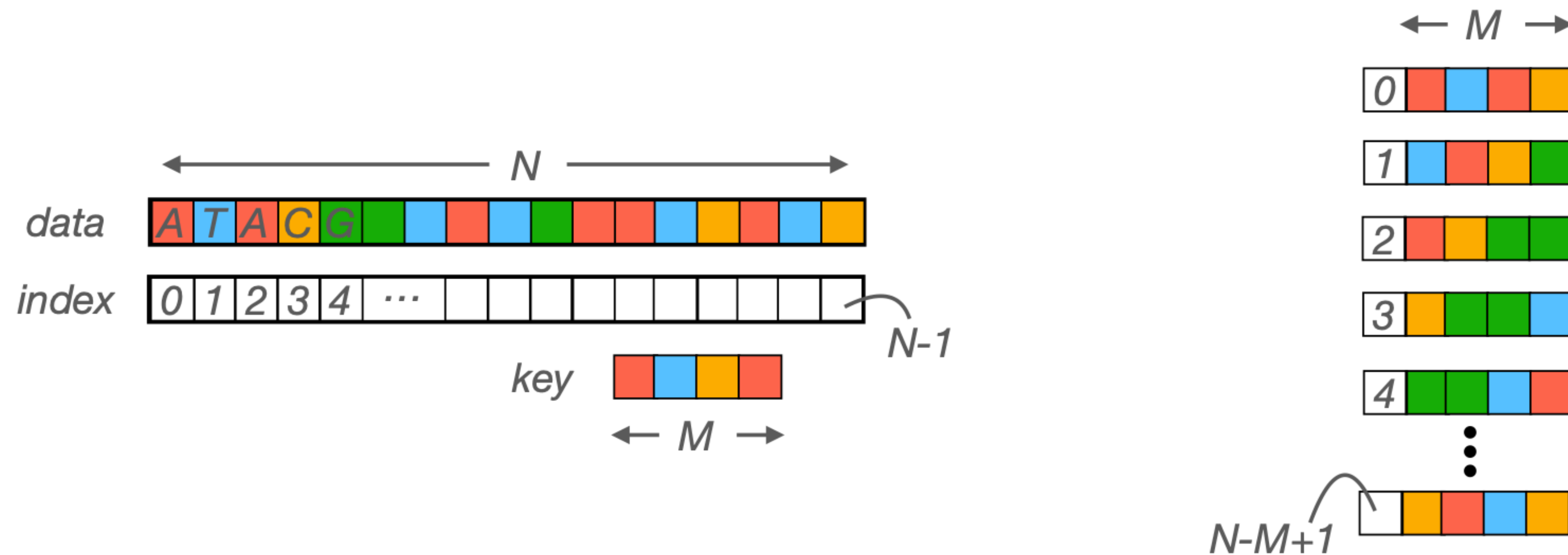
Grover's
circuit



Problem: the
data loading
circuit already
require depth N

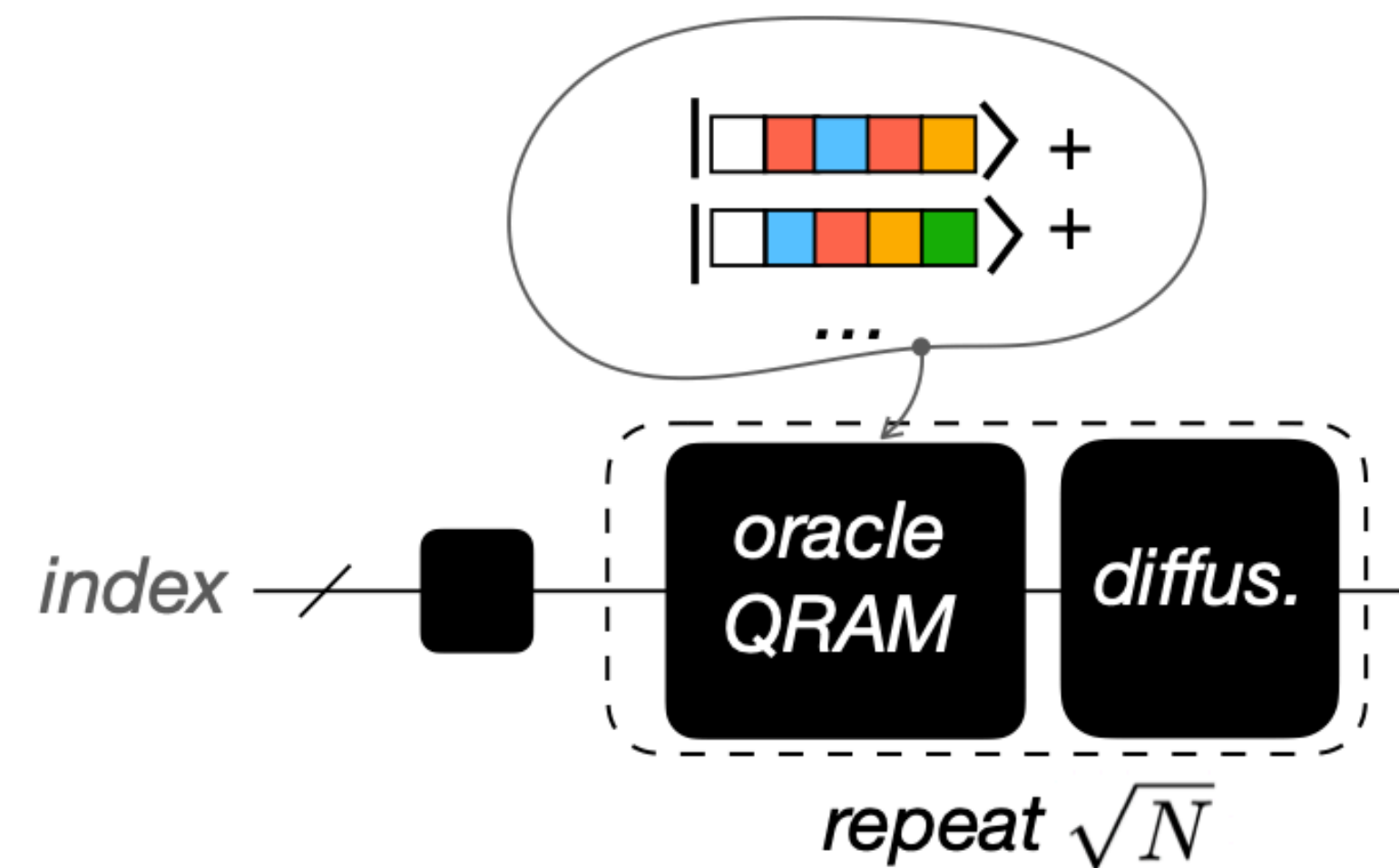
Input model: an example

genome



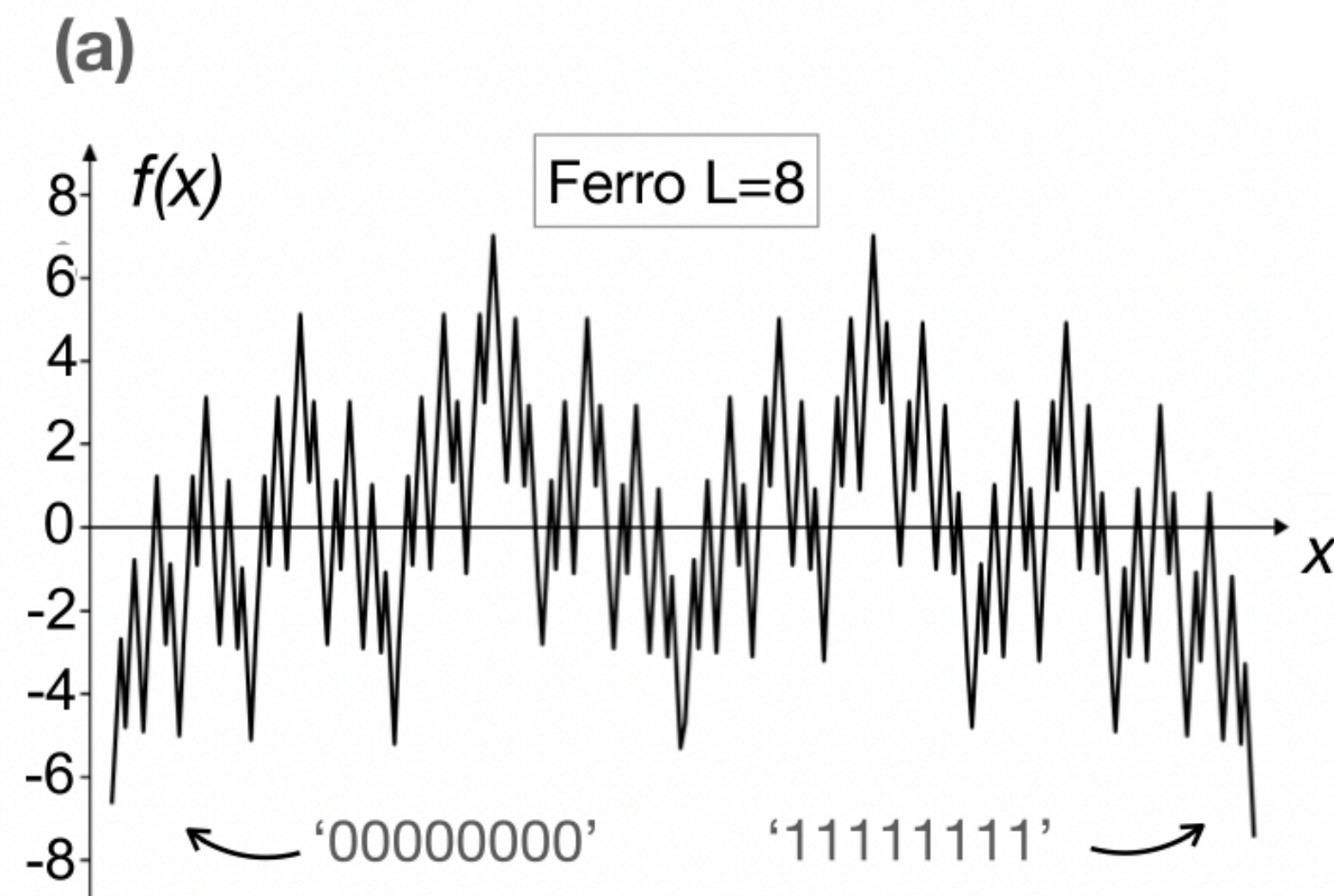
Transformed
into a database
"phonebook"

Grover's
circuit
assuming a
QRAM



Output model: VQE or QAOA

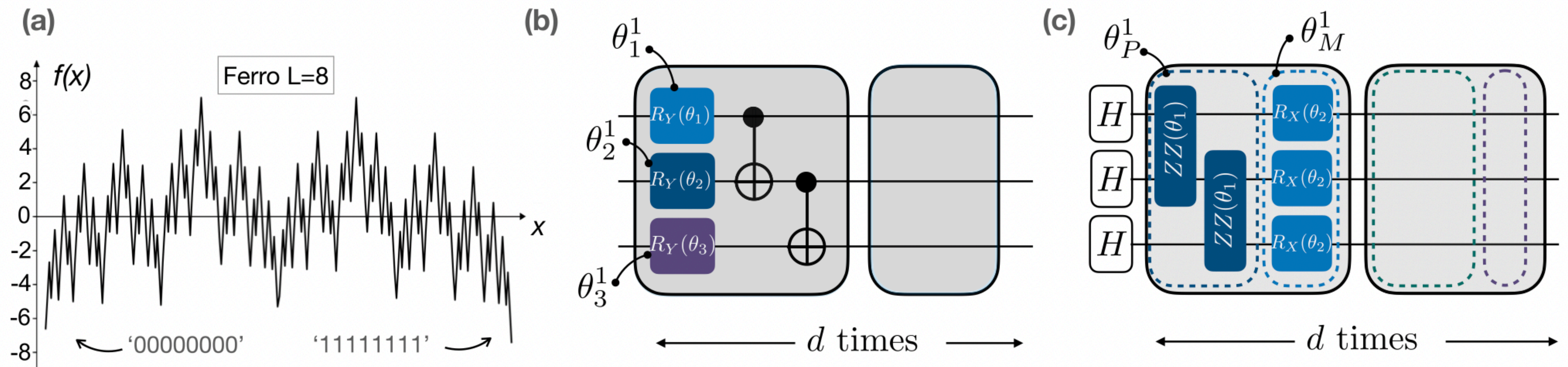
Task: Finding ground-state of the (classical) Hamiltonian $\hat{H}_P = \sum_x f(x) |x\rangle \langle x|$



Output model: VQE or QAOA

Task: Finding ground-state of the (classical) Hamiltonian $\hat{H}_P = \sum_x f(x) |x\rangle \langle x|$

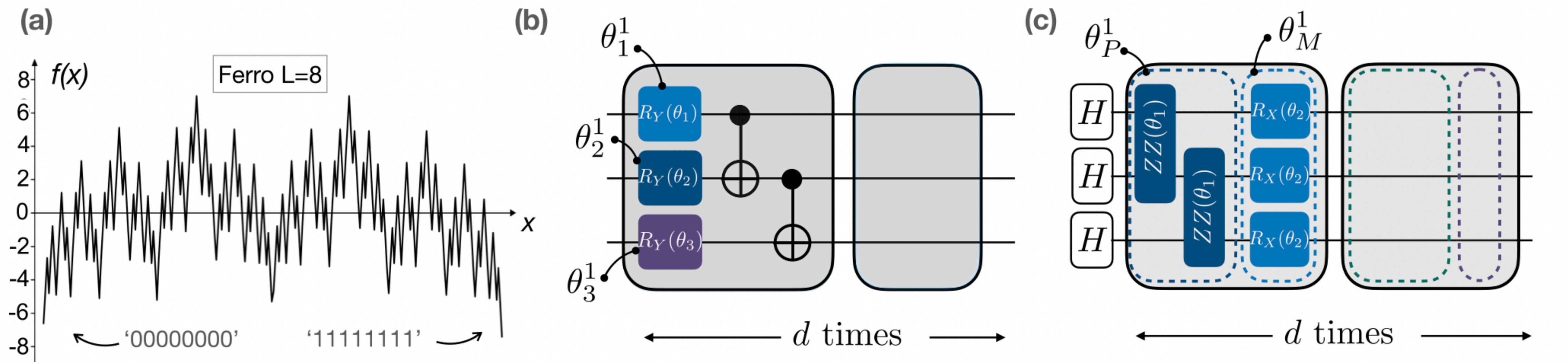
How: Using a parametrised circuit, and optimising it following a cost function (or gradients)



Output model: VQE or QAOA

Task: Finding ground-state of the (classical) Hamiltonian $\hat{H}_P = \sum_x f(x) |x\rangle \langle x|$

How: Using a parametrised circuit, and optimising it following a cost function (or gradients)



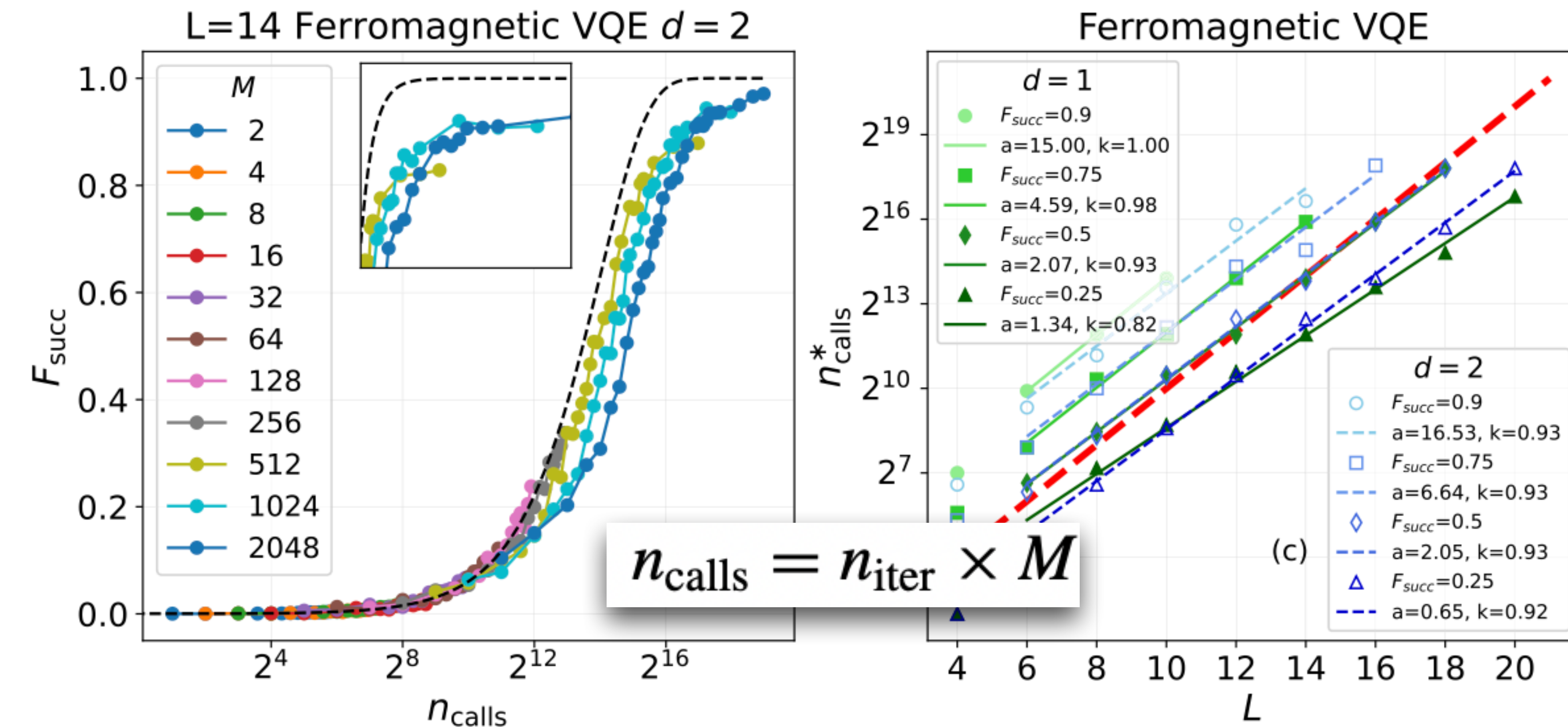
Issue: Cost function is evaluated from M wf collapses, each one requires a circuit execution

$$\tilde{C} = \langle \psi_{\theta} | \hat{H}_P | \psi_{\theta} \rangle = \sum_{x=0}^{2^L-1} |\psi_{\theta}(x)|^2 f(x) \rightsquigarrow \tilde{C} \approx \frac{1}{M} \sum_{i=1}^M f(x_i), \quad \text{where } x_i \text{ are sampled from } |\psi_{\theta}(x)|^2$$

Output model: VQE or QAOA

What is the real scaling of VQE and QAOA, counting also the shots?

Here we count resources as “calls” to the classical function $f(x)$.. where x is the bit-string read out of each measure



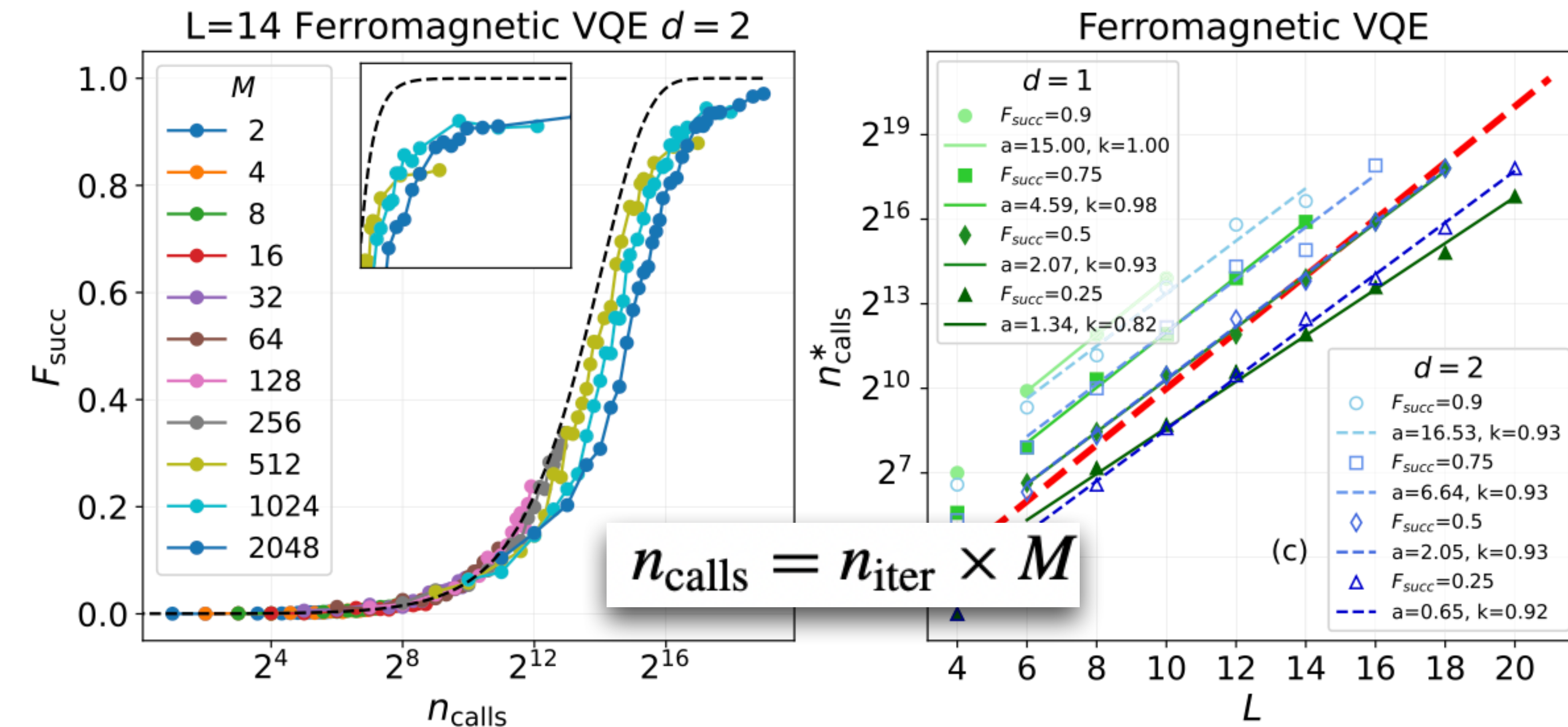
We use different combinations of M shots and n_{iter} iterations.

From initial random parameters

Output model: VQE or QAOA

What is the real scaling of VQE and QAOA, counting also the shots?

Here we count resources as “calls” to the classical function $f(x)$.. where x is the bit-string read out of each measure



We use different combinations of M shots and n_{iter} iterations.

From initial random parameters

Results

Number of shots to reach a target success probability

$$n \sim 2^{kL}$$

VQE energy based $k \approx 0.9$

VQE gradient based $k \approx 0.5$

QAOA best setup $k \approx 0.3$

Good candidates for quantum advantage?

- Small input
- Output is simple to read

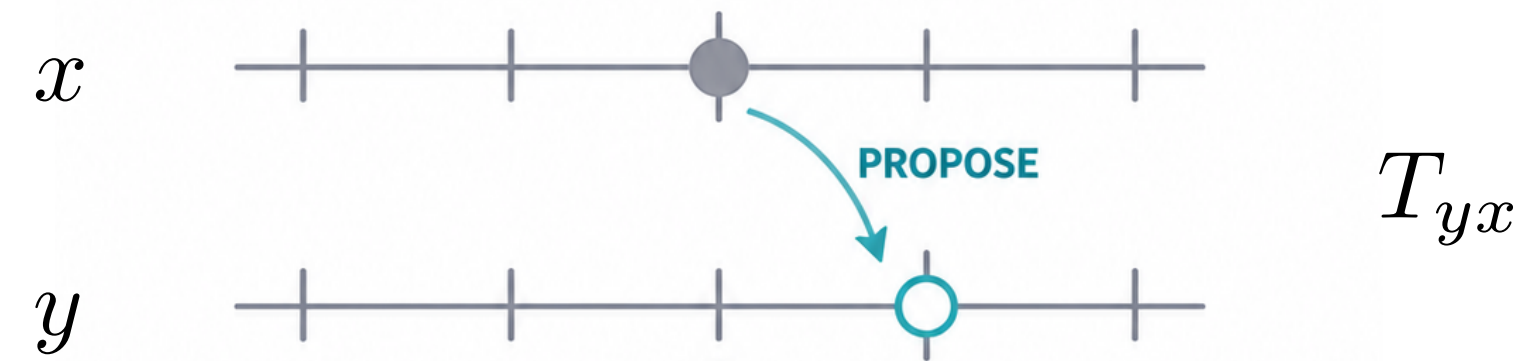
Additional features:

- Result verifiable
- Classical benchmark well established
- Not classically simulatable / dequantizable

The Metropolis Algorithm

Goal: Sample from an equilibrium distribution: $\pi_\beta \propto e^{-\beta H(x)}$

The Metropolis Algorithm



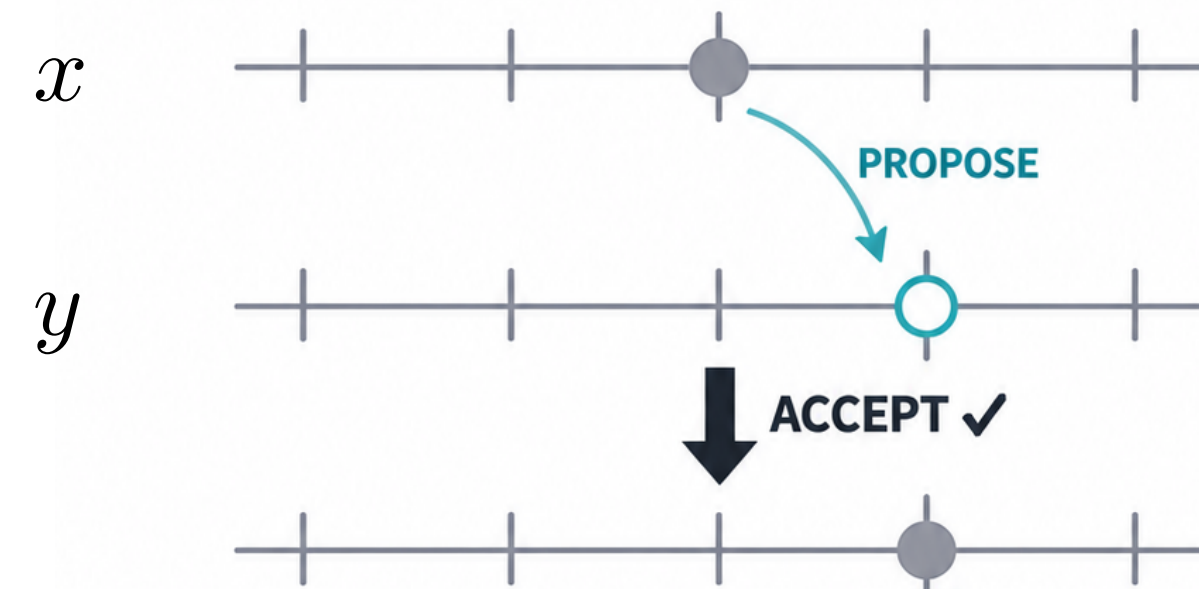
Goal:

Sample from an equilibrium distribution: $\pi_\beta \propto e^{-\beta H(x)}$

How:

Transition probability from x to y : $P_{yx} = T_{yx} A_{xy}$

The Metropolis Algorithm



T_{yx}

A_{yx}

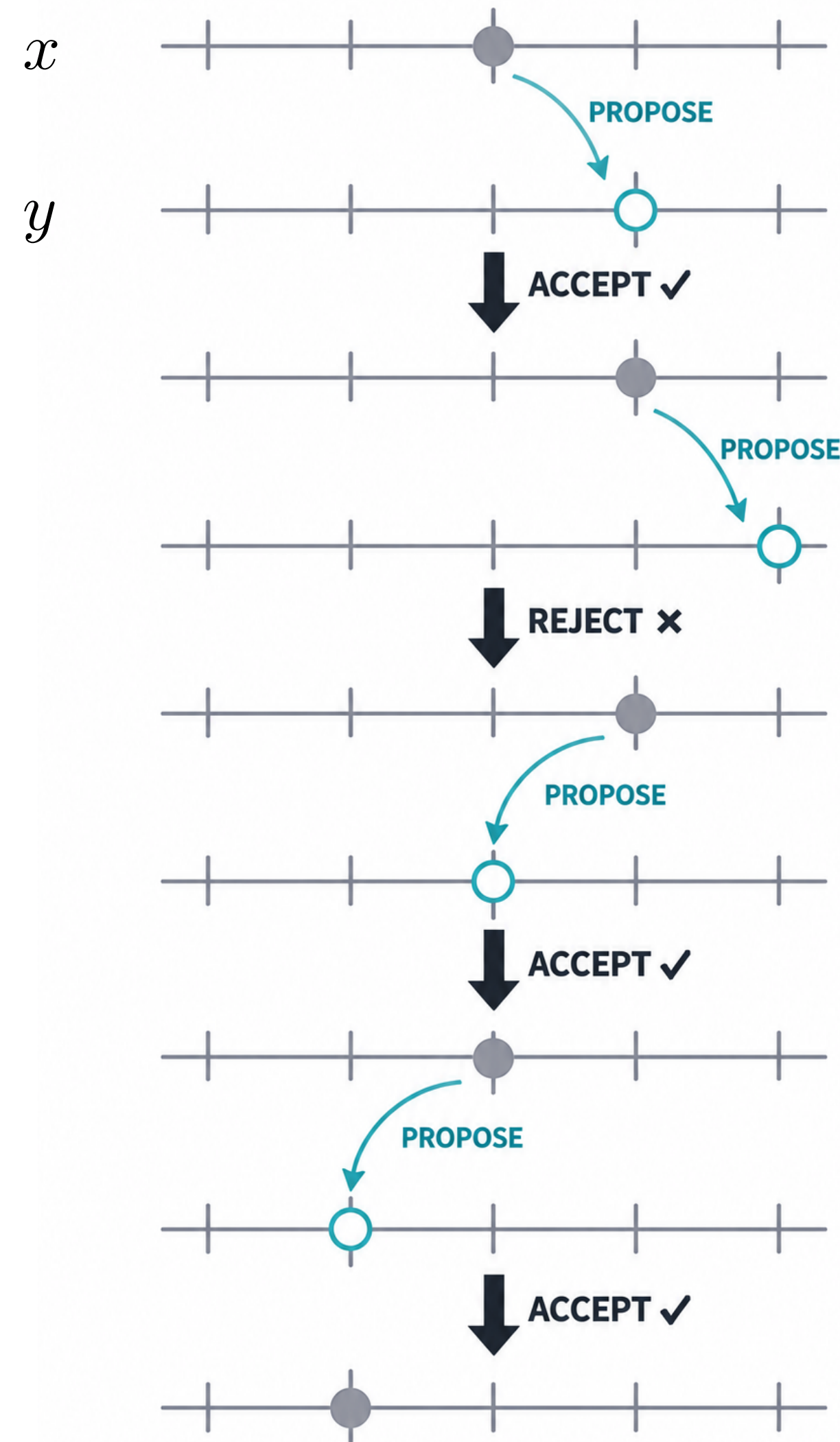
Goal:

Sample from an equilibrium distribution: $\pi_\beta \propto e^{-\beta H(x)}$

How:

Transition probability from x to y : $P_{yx} = T_{yx}A_{xy}$

The Metropolis Algorithm



T_{yx}

A_{yx}

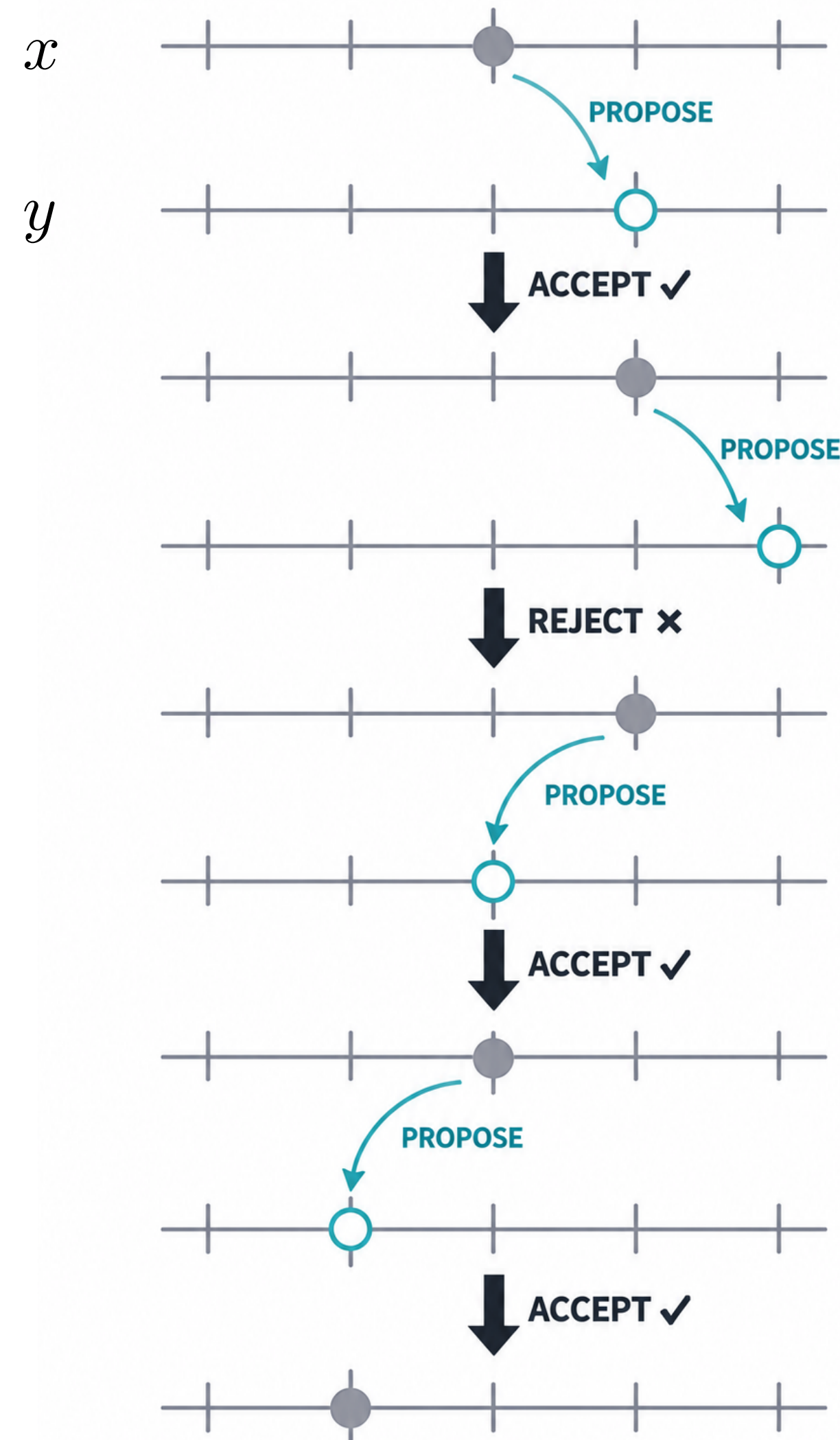
Goal:

Sample from an equilibrium distribution: $\pi_\beta \propto e^{-\beta H(x)}$

How:

Transition probability from x to y : $P_{yx} = T_{yx}A_{xy}$

The Metropolis Algorithm



$$T_{yx}$$

$$A_{yx}$$

Goal:

Sample from an equilibrium distribution: $\pi_\beta \propto e^{-\beta H(x)}$

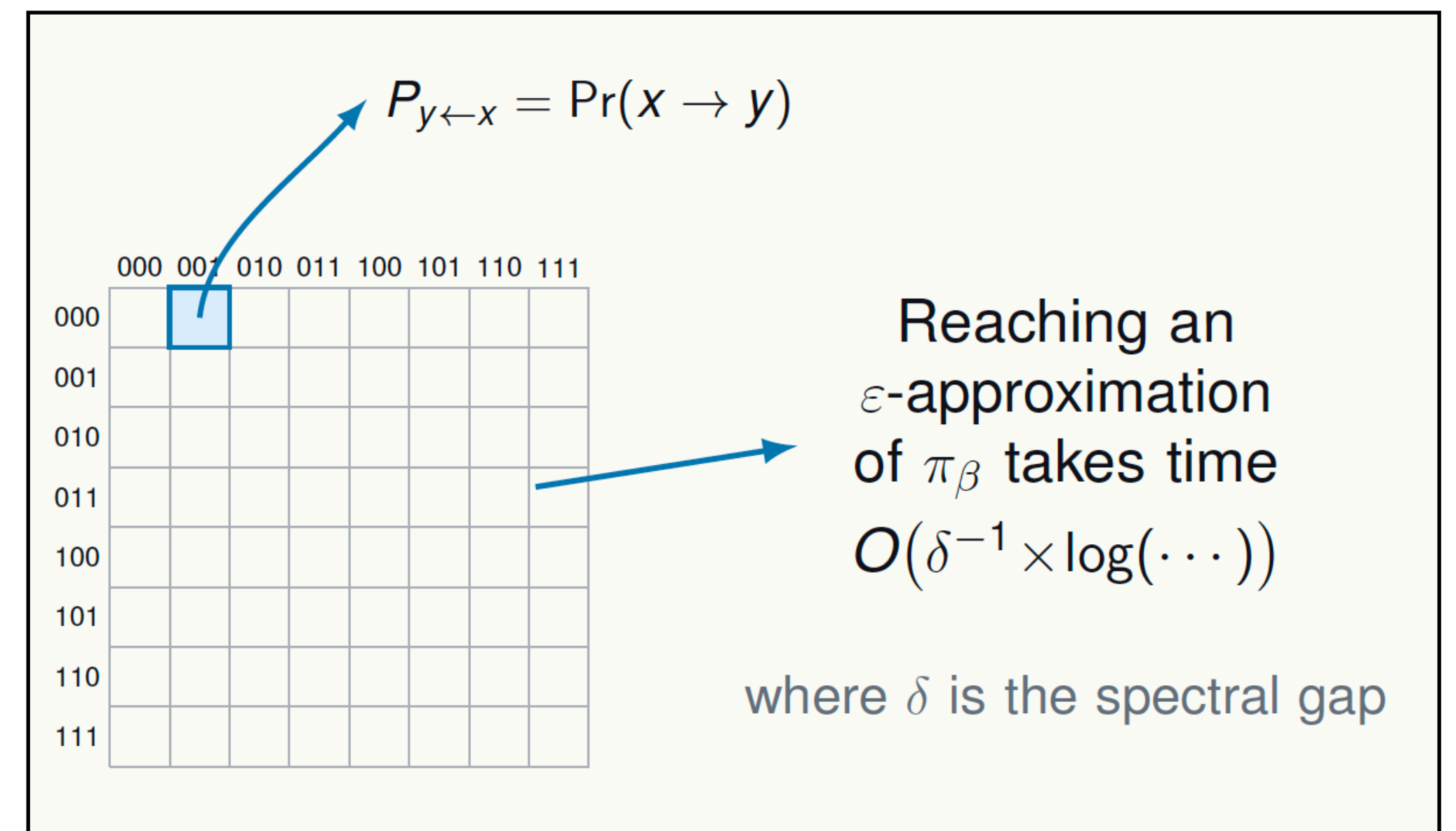
How:

Transition probability from x to y: $P_{yx} = T_{yx}A_{xy}$

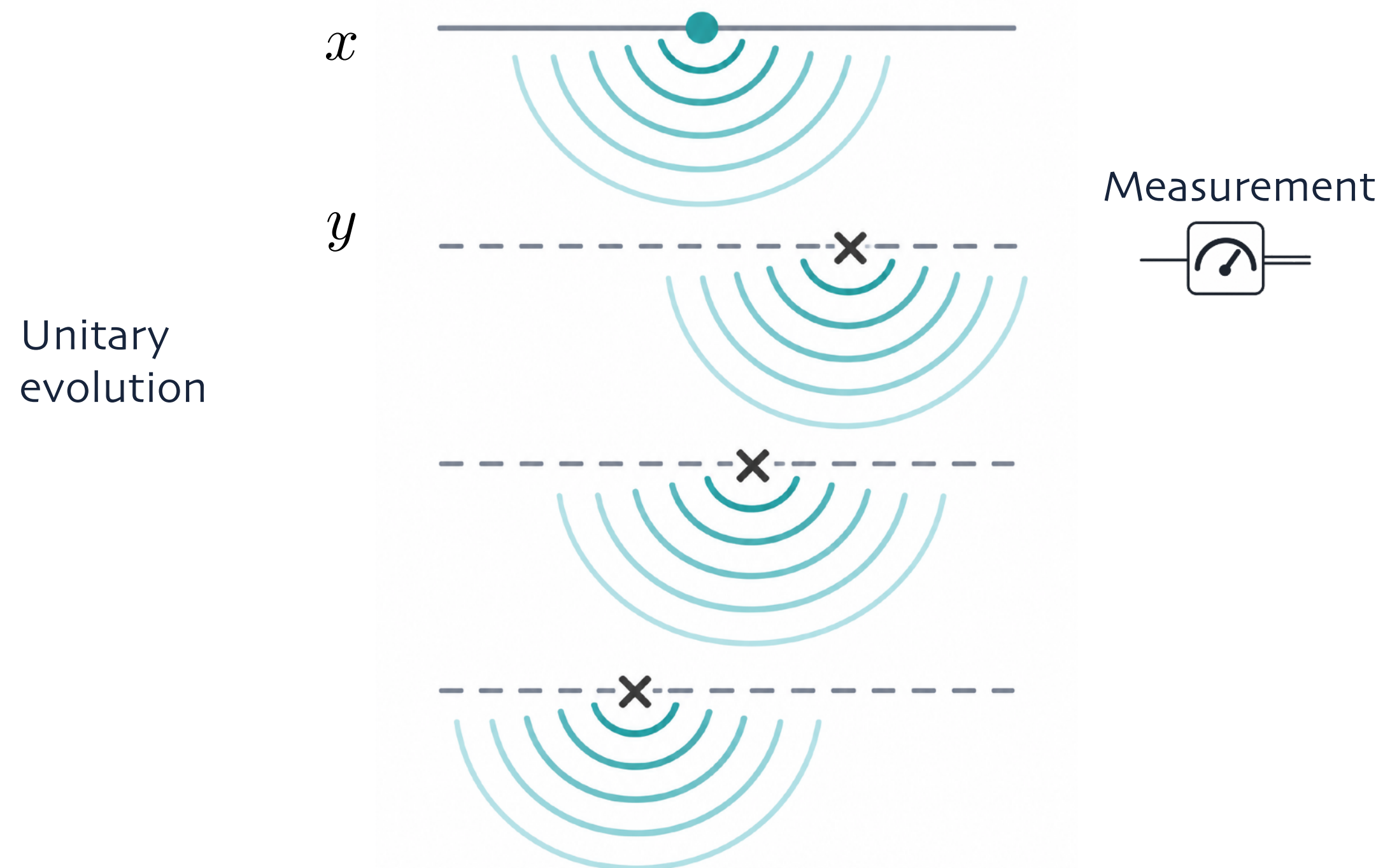
Problem:

The chain of configurations $\{x_0, x_1, \dots, x_N\}$ may not “explore” efficiently the whole space, in N steps

Measuring the efficiency of a Markov chain



The Quantum-enhanced Metropolis Algorithm



- Proposal move is done on quantum hardware

$$T_{yx} = |\langle y | U(t, \gamma) | x \rangle|^2$$

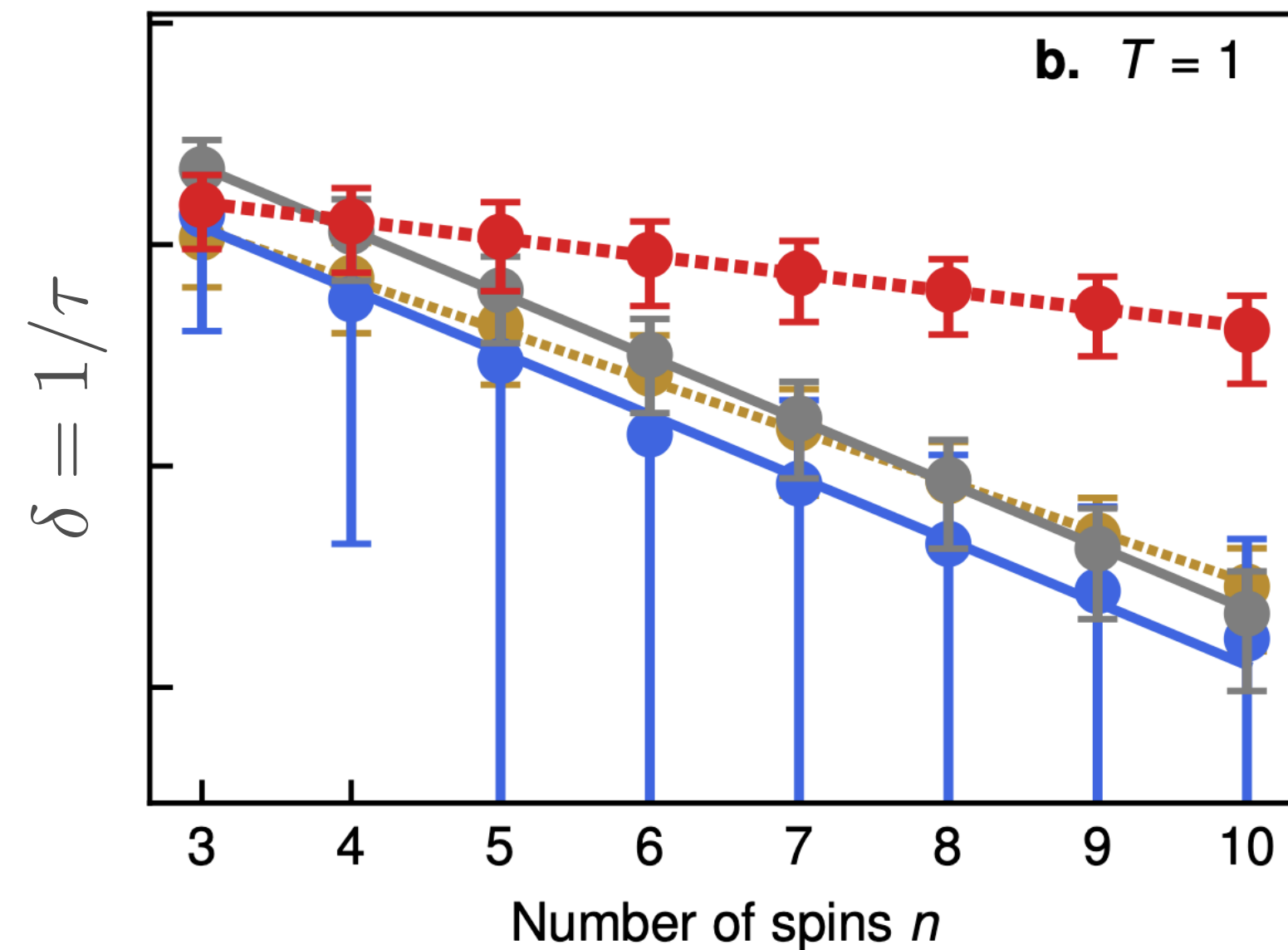
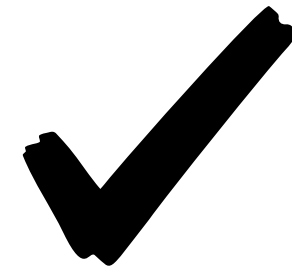
$$U(t, \gamma) = \exp(-itH_{\text{tf}}(\gamma))$$

- Measurement selects just one trial conf.
- Acceptance A_{yx} done classically

Mazzola, **Phys Rev A**, 104, 022431 (2021)

Layden, Mazzola, et. al. **Nature**, 619, 282 (2023)

The Quantum-enhanced Metropolis Algorithm



c. $\langle \delta \rangle \propto 2^{-kn}$ fits

Quantum	$k = 0.264(4)$
Local	$k = 0.94(4)$

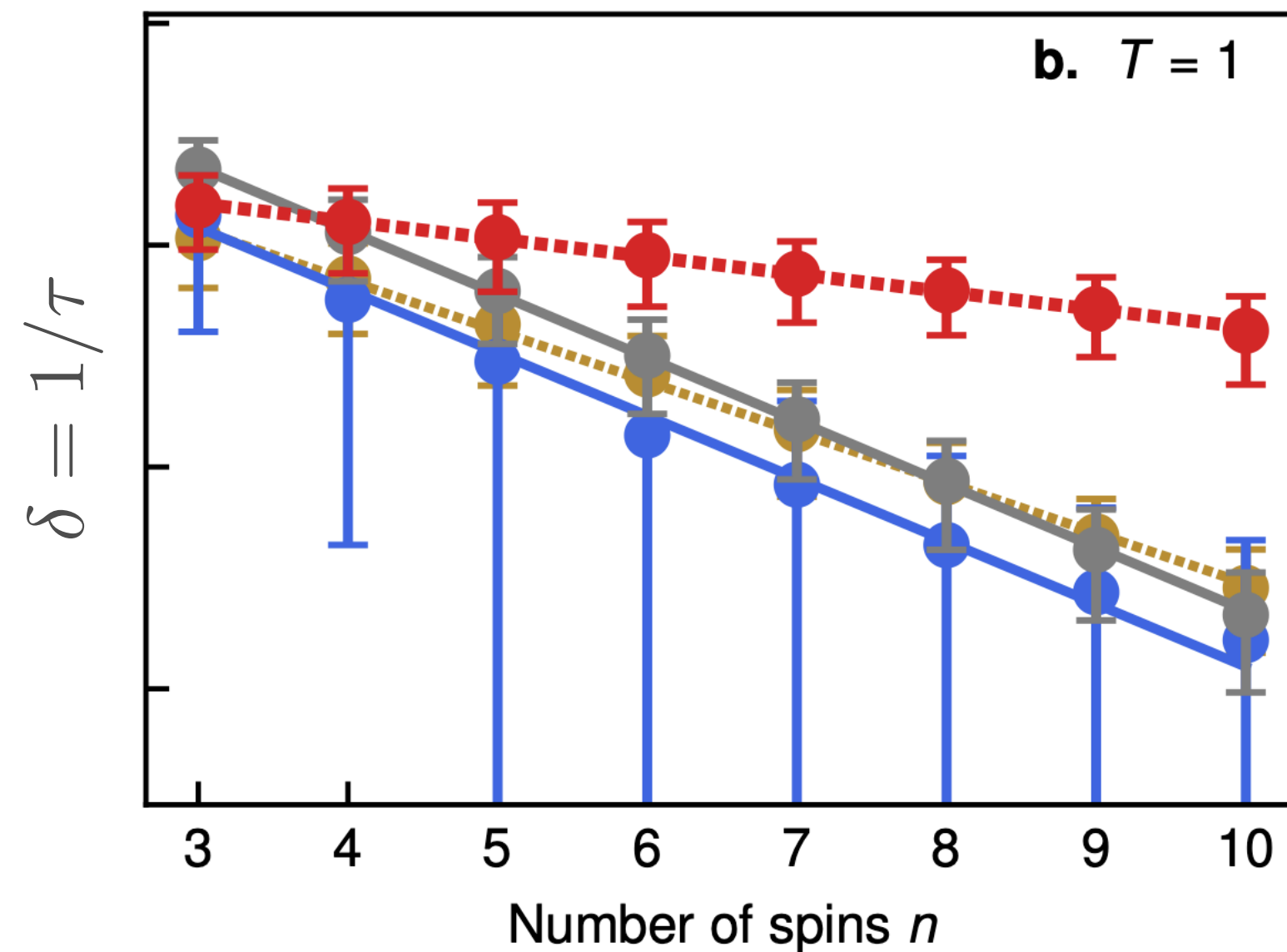
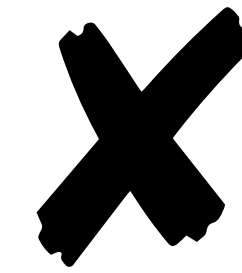
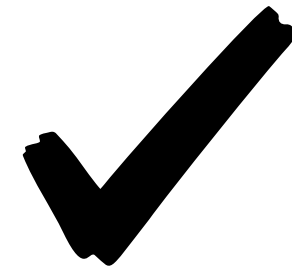
Quantum enhancement factor in k : 3.6(1)

Empirical cubic scaling advantage!

Mazzola, **Phys Rev A**, 104, 022431 (2021)

Layden, Mazzola, et. al. **Nature**, 619, 282 (2023)

The Quantum-enhanced Metropolis Algorithm

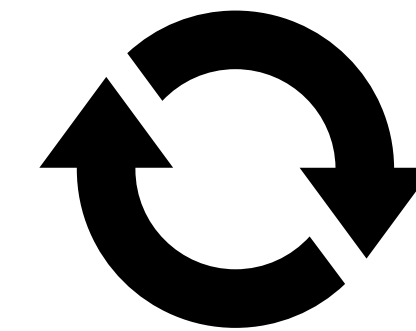
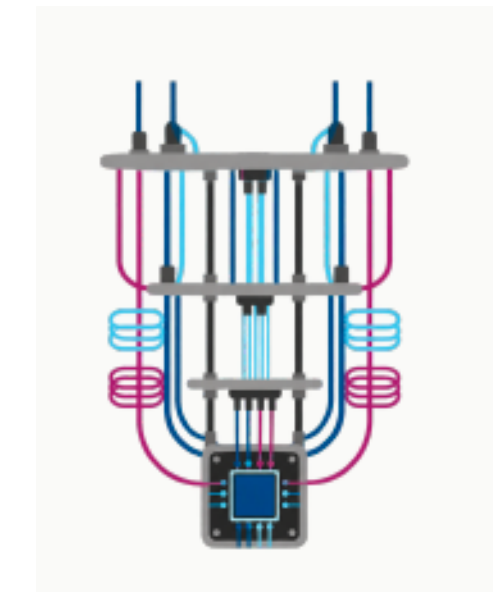


c. $\langle \delta \rangle \propto 2^{-kn}$ fits

Quantum	$k = 0.264(4)$
Local	$k = 0.94(4)$

Quantum enhancement factor in k : 3.6(1)

Empirical cubic scaling advantage!



- Large communication overhead
- Short depth is good for NISQ but can we do more fully coherently?

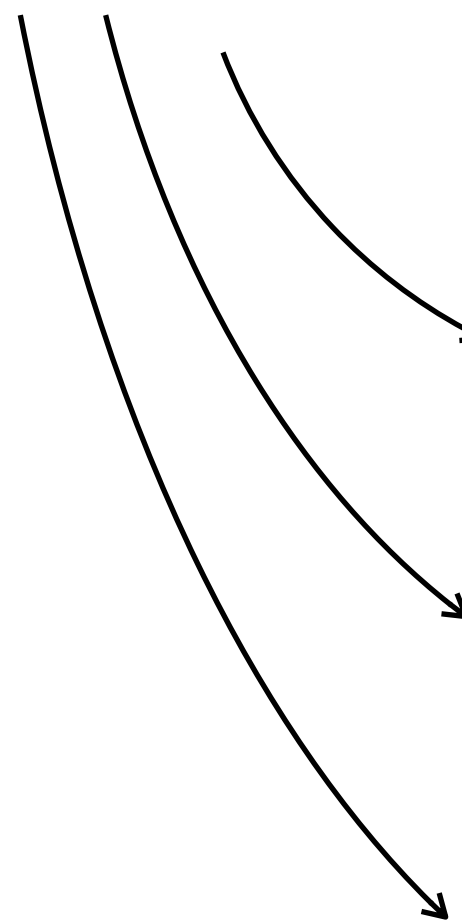
Mazzola, **Phys Rev A**, 104, 022431 (2021)

Layden, Mazzola, et. al. **Nature**, 619, 282 (2023)

Quantum walks

(Szegedy's) Quantum-walks

$$W = R_0 V^\dagger B^\dagger F B V,$$


$$V |x\rangle_a |0\rangle_b = |x\rangle_a \sum \sqrt{T_{yx}} |y\rangle_b$$

$$B |x\rangle_a |y\rangle_b |0\rangle_c = |x\rangle_a |y\rangle_b \left(\sqrt{1 - A_{yx}} |0\rangle_c + \sqrt{A_{yx}} |1\rangle_c \right)$$

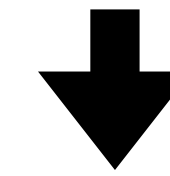
F swaps a with b when the coin flag C is $|1\rangle_c$

Promise: a quadratic improvement in the spectral gap: $\delta \rightarrow \sqrt{\delta}$

Fully-Quantum walks

Quantum walks exists since more than 20 years...

...but always “quantizing” classical trial moves, i.e. spin-flips

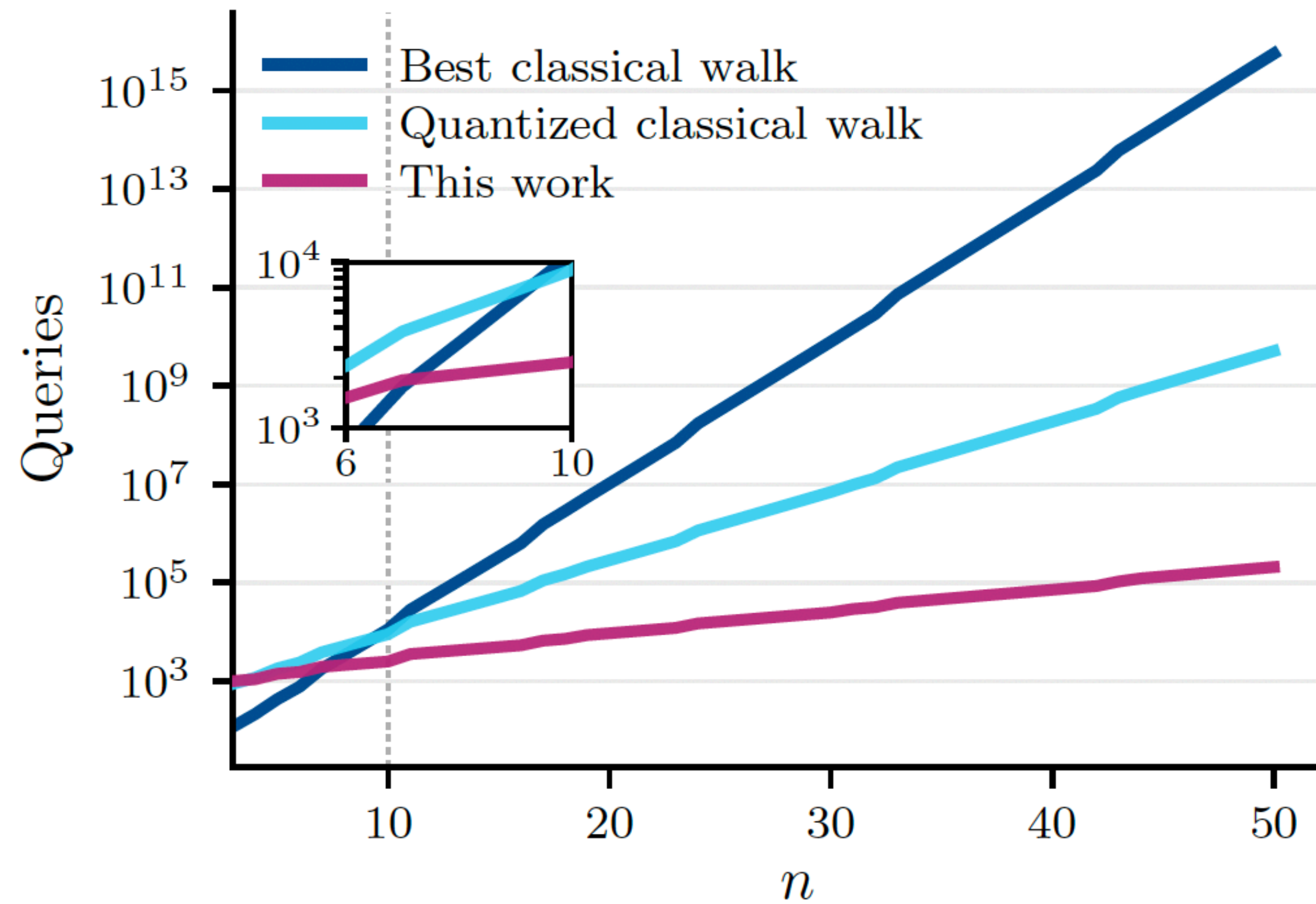


$$V |x\rangle_a |0\rangle_b = |x\rangle_a \sum_y \sqrt{T_{yx}} |y\rangle_b$$

What if we use our “quantum-enhanced” proposal dynamics $\propto T_{yx}$?

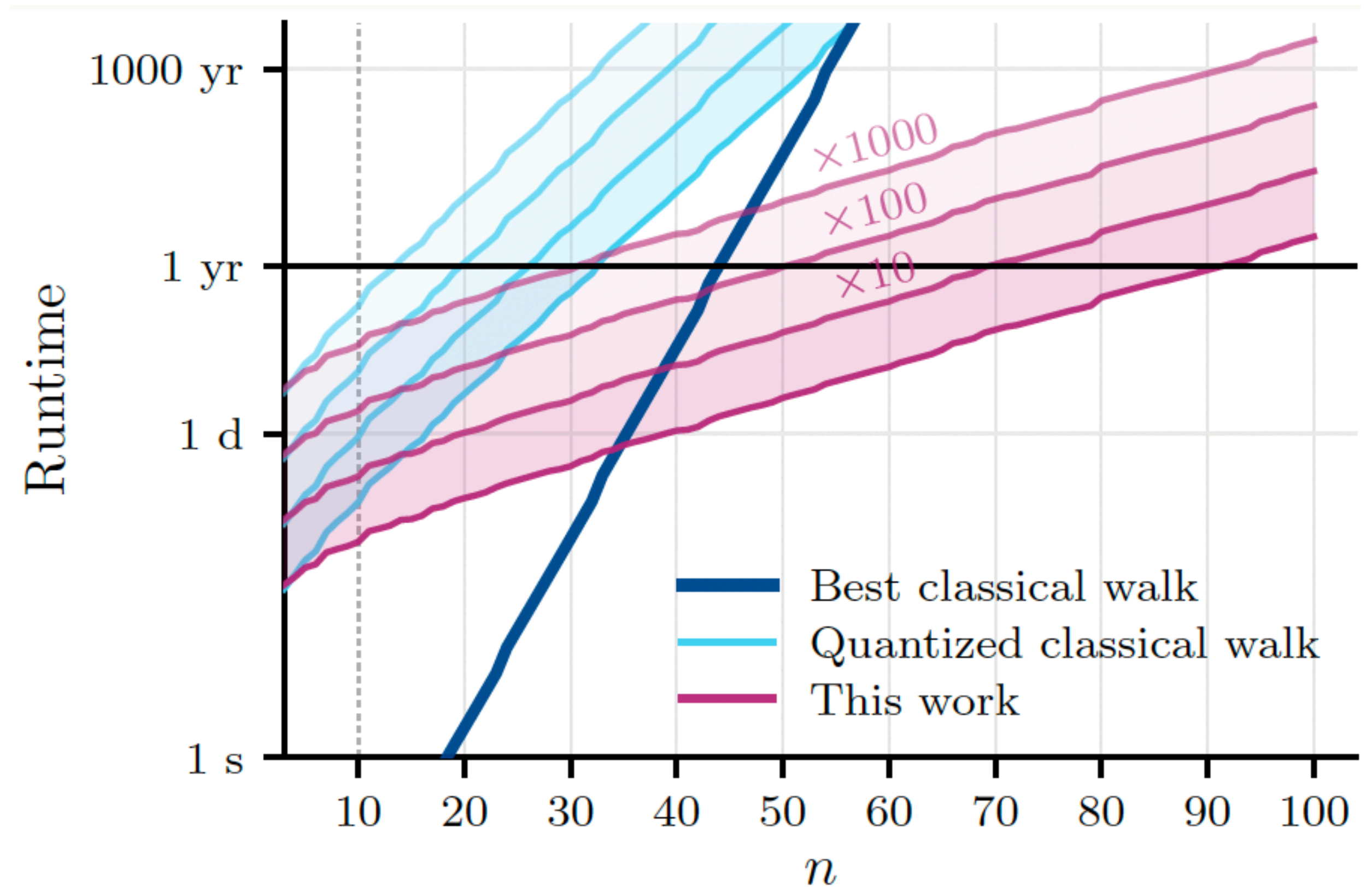
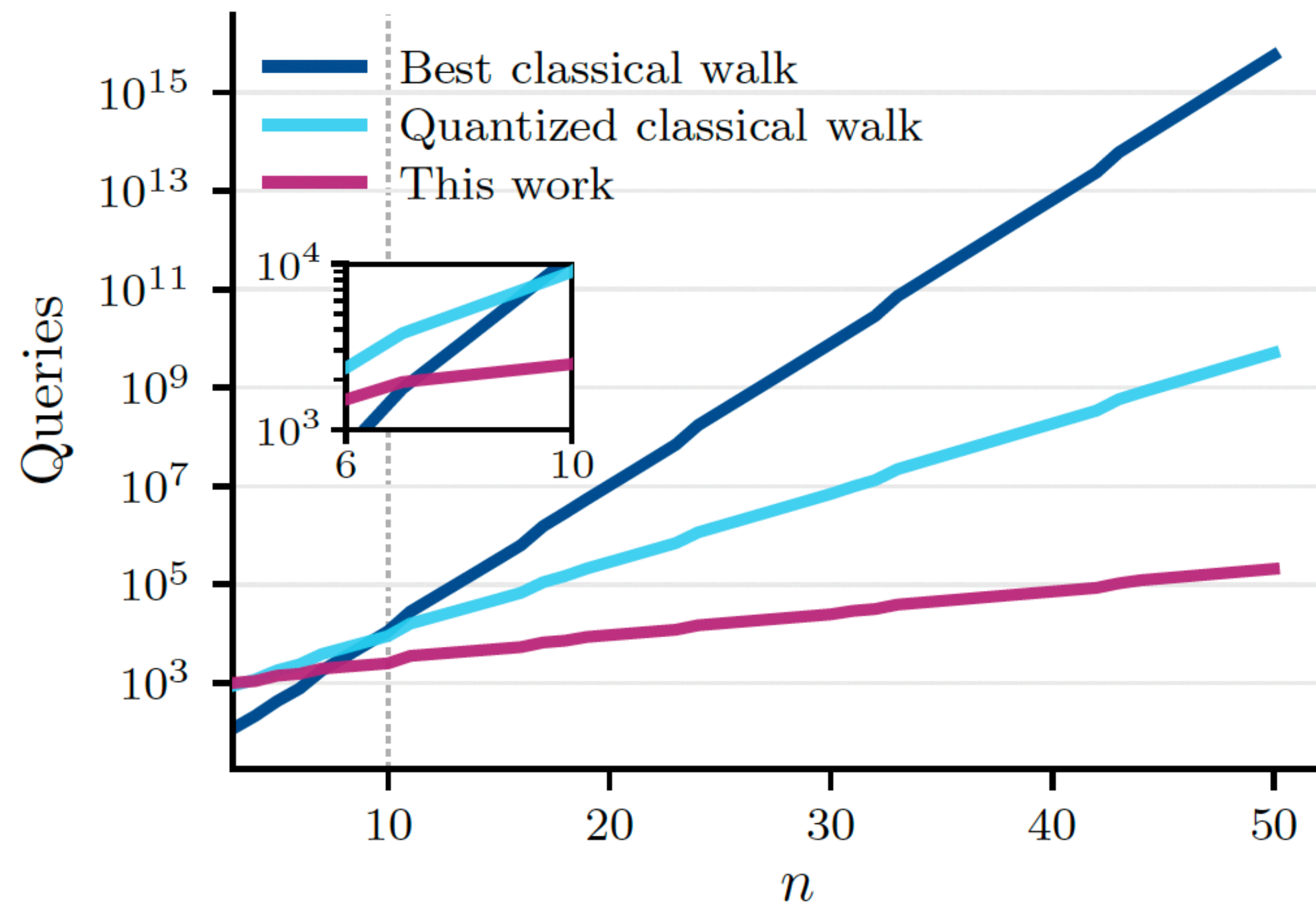
Fully-Quantum walks

Incudini & Mazzola, Practical advantage beyond the quadratic speedup limit with fully-quantum walks, arXiv:2607.22818



Fully-Quantum walks

Incudini & Mazzola, Practical advantage beyond the quadratic speedup limit with fully-quantum walks, arXiv:2607.22818



Compiled using Clifford + T + rotations

Surface code, with physical op time from 20 ns to 20 μ s

Ref: Beverland et al., arXiv:2211.07629 (2022);
Litinski, Quantum 3 (2019); Google Quantum AI, Nature 638 (2025).

Thank you!